

ALPHA-HP-200 Hydraulic Press - Energy Isolation and Zero-Energy Verification

Field	Value
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Tenant	Alpha Manufacturing (tenant-alpha)
Site	Ridgeway Works, Building 2 (metric documentation)
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Source identifier	AM-DOC-1071
Equipment	ALPHA-HP-200
Also covers	ALPHA-HP-200B
Document type	Safety / energy isolation procedure
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Applies to	Assets listed under equipment
Related documents	ALPHA-MAN-HP200-B, ALPHA-PM-HP200-B, ALPHA-DIA-P100-1
Document owner	EHS and Maintenance Engineering

Purpose and scope

This procedure describes how electrical, hydraulic, gravitational and residual energy are made safe on the ALPHA-HP-200 press and on its power unit ALPHA-P-100 before any person opens a guard, touches a tie rod, handles a hose or fitting, opens the manifold, works on the die height sensor or performs any diagnostic measurement inside the die area.

Applies to ALPHA-HP-200 and ALPHA-HP-200B. It does not apply to ALPHA-HP-210, which is isolated under AM-SAF-HP210-LOTO-1.

Rule 1. No person may open a guard, break a seal, touch a hydraulic line or enter the die area of this press unless this procedure has been completed for that specific task and the green zero-energy verification tag is attached to the isolation points.

Rule 2. Rule 1 applies even when the press appears switched off. The ram, the counterbalance cylinders and accumulator AC-1 store energy after the pump motor stops.

Energy sources on this machine

Energy	Source	Isolation point
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Electrical	400 V three phase	Feeder AM-MCC-3 Panel 3A-04, breaker 3A-04-01
Electrical	control 24 V DC	Control transformer in panel 3A-04 Breaker 3A-04-07
Hydraulic	pressurised	Power unit ALPHA-P-100 Ball valves HV-101 and HV-201
Stored hydraulic pressure	Accumulator AC-1, nitrogen pre-charge	Manual bleed valve BV-103 into drain vessel DV-1
Gravitational	Ram mass, about 1600 kg	Mechanical block plus tie-off on the ram yoke
Pneumatic	control only	6 bar instrument air Ball valve AV-311
Residual	Pressurised hoses that have been disconnected	Cap after disconnection, blank plug before work

The press has no pneumatic energy in the press circuit itself. Instrument air actuates the clamp circuit only.

Required sequence

Perform these six steps in order. Skipping a step is an incident and must be reported.

1. Notify. Inform the Bay 2 shift supervisor and the operators of press line 2 that the press will be out of service, and record the notification in the permit PTW-2600.
2. Stop normally. Press the stop button at the pendant, let the pump motor run down for the full 40 seconds, then set the controls to neutral. Do not use the emergency stop as a normal stopping method; the emergency stop drops the ram in an uncontrolled way.
3. Isolate electrical energy. Open panel 3A-04, set breaker 3A-04-01 to OFF and the isolator handle to the locked position. Fit your personal lock and a tag. Fit the lock box if more than one person works on the press. Verify the control supply at breaker 3A-04-07 is also isolated. A padlock on the isolator handle is the only accepted electrical isolation for this press.
4. Isolate the hydraulic supply. Close ball valve HV-101 (supply from the pump) and ball valve HV-201 (return to tank). Both valves have padlockable brackets. Fit locks and tags. Close instrument air at AV-311.
5. Dissipate stored energy. This step is not optional and cannot be delegated.
 - Open manual bleed valve BV-103 and discharge accumulator AC-1 into drain vessel DV-1. Never vent a charged accumulator to atmosphere.
 - Watch the accumulator gauge on power unit ALPHA-P-100. Continue bleeding until the gauge reads 5 bar or less. If it will not come below 5 bar, stop and escalate; the accumulator or the gauge is defective and the circuit must be treated as charged.
 - Run the pump motor down in bypass if the pump has not been run down naturally, then close HV-101.
 - Fit the mechanical block under the ram yoke and fit the tie-off on the ram yoke. The block and the tie-off must both be fitted before hands enter the die area.
6. Verify the zero-energy state. Perform all four verifications:
 - Attempt start: try to start the press from the pendant. Nothing must move and the pendant must show the lockout state.
 - Attempt to operate the isolation valves: the locked ball valves must not move.
 - Attempt to move the ram by hand at the ram crown. It must not move.
 - Confirm the accumulator gauge reads 5 bar or less and the pressure gauge at the manifold reads 0 bar.Attach the green zero-energy verification tag. The tag is signed by the technician and countersigned by a second authorised person who has personally witnessed step 6.

A document-controlled attempt start verification is mandatory for this press. The attempt start record is part of permit PTW-2600 and is audited.

Rules that are frequently broken

Working on a hose or fitting because "the pump is off". The pump being off is not isolation. HV-101, HV-201 and the accumulator bleed are all required. Assuming the accumulator is empty because it was bled last week. The gauge

reading is the evidence, on the day, for that person.

Removing a hose and leaving the port open. Cap the port immediately and install a blank plug before the next task.

Using a hoist, a forklift or a vehicle as a test weight or as a hold-down.

Leaving a personal lock off the isolator when the work stops for a break.

Every break means re-verification.

Defeating the two-hand control to speed up a mechanical job. This requires a documented risk assessment, a permit and a second authorised person.

Return to service

- 1. Remove tools, blanks and test pieces from the die area. Fit all guards and fix all fasteners. Check that the tie-off and the mechanical block are still serviceable and return them to the store.
- 2. Close all bleed and drain points that were opened, and remove drain vessel DV-1.
- 3. Remove locks one at a time. Each person removes their own lock. The person who applied the isolator supervises the removal.
- 4. Tell the Bay 2 supervisor that the press is released.
- 5. Restore panel 3A-04 to ON and close HV-101, HV-201 and AV-311.
- 6. Recharge accumulator AC-1 through the charging point. The pre-charge value is the as-built value on the data plate; do not charge from a figure copied from another press.
- 7. Restore electrical energy and run a dry cycle with the die area empty and hands clear of the die area. Confirm the ram raises and lowers normally, the pressure reading is steady and no leak is visible.
- 8. Record the return to service in permit PTW-2600 and on the work order.

If the work involved the tie rods, the bolsters, the ram or the frame, the post-run checks in AM-PM-HP200 apply before production use.

Incident reporting

Any of the following is a reportable event and is reported to EHS before the next shift, even if nobody was hurt:

- A guard was opened or a fitting was loosened while the circuit was charged.
- An attempt start produced any movement of the ram.
- Stored energy was released onto a person, or a person entered the die area with the ram loaded.
- A lock was found unlocked, or a tag was found without a lock.
- A person worked on the press without a completed permit PTW-2600.

Appendix: Energy isolation point schedule

The energy isolation points for the ALPHA-HP-200 press and its hydraulic power unit are listed below. Every point is locked and tagged with the green zero-energy tag before work begins. The schedule is verified against the as-built drawing at each annual review.

Ref	Energy source	Device	Location	Method	Verification
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HV-101	Electrical, main	Isolator	Panel 3A-04	Lock open	Test for dead at motor terminals
HV-201	Electrical, control	Isolator	Panel 3A-04	Lock open	Test control voltage absent
HY-1	Hydraulic, stored	Relief and drain valve	Manifold	Open and drain	Gauge reads zero at TP-3

AC-1	Pneumatic, accumulator	Block valve BV-103	Manifold	Open to drain vessel	Accumulator gauge reads zero
ME-1	Mechanical, gravity	Ram block and tie-off	Frame	Fit mechanical stop	Ram cannot move under load
TH-1	Thermal	Oil cooler	Power unit	Allow to cool	Surface temperature below 40 degC

Isolation is not complete until every point in the schedule has been applied and the zero-energy state has been verified by attempting to start the press and confirming that nothing moves. The verification attempt is recorded with the date, time and the name of the person who made it.

Two people are required for verification: the person who applied the isolation and a second person who confirms the zero-energy state independently.

Appendix: LOTO permit and record retention

The isolation permit PTW-2600 is raised for each job and closed only after the isolation has been removed and the press has been returned to service. The permit identifies the asset, the work order, the isolation points applied, the verifying person and the time of removal.

Records are retained for seven years. The record set for each job contains:

the raised permit and the lock and tag numbers issued;
the zero-energy verification record with the attempted-start result;
the list of isolation points and how each was proven dead;
the name of the person who applied the isolation and the verifier;
the time the isolation was removed and the press returned to service;
any deviation, and the approval that authorized it.

A permit that cannot be produced when requested is treated as an unverified isolation and the work is stopped until the record is reconstructed and re-verified.

Revision history

Revision	Effective date	Summary
Rev 1	2023-05-15	First controlled issue for the four-post press and its power unit.